

1. Introduction:

In this work I implemented a gait-based biometric system that can both identify and verify users from smartphone sensor data. The dataset contained accelerometer and gyroscope readings from ten different users. The goal was to preprocess the raw signals and train neural-network models that recognize each person's walking pattern. I first cleaned and prepared the data, then built two types of models:

- Feedforward Neural Network (FNN) using handcrafted features and
- Long Short-Term Memory (LSTM) network that learns directly from the raw time-series.

Both architectures were evaluated for identification (multi-class) and verification (one-vs-all binary) tasks.

2. Methodology

2.1 Preprocessing

Each user had two CSV files, one from the accelerometer and one from the gyroscope. I merged them by timestamp into a six-channel signal:

[Acc_X, Acc_Y, Acc_Z, Gyro_X, Gyro_Y, Gyro_Z].

The continuous data was segmented into 5-second windows with 50 % overlap (≈ 250 samples $\times$ 6 features per window at 50 Hz). Noise analysis showed random spikes and small vibrations, so I used Z-score outlier clipping ($|z| > 3$) followed by a 5-point Moving Average filter.

Justification

I split the continuous data into 5-second windows with 50% overlap to capture several walking cycles while keeping the signal stable and increasing sample count. After merging accelerometer and gyroscope readings, I noticed spikes and small jitters. To clean them, I used Z-score clipping ($|z| > 3$) and a 5-point moving-average filter. This simple combination removed extreme noise and smoothed high-frequency jitter while keeping the main gait pattern intact.

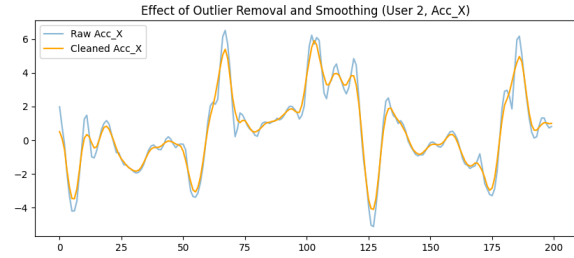


Figure 1: Raw and Cleaned Signal Example

2.2 Feedforward Neural Network (FNN)

Feature Extraction and Selection

From each 5-second window I extracted 78 statistical and frequency features (13 per channel: mean, std, variance, skewness, kurtosis, min, max, IQR, MAD, FFT mean/std, dominant frequency, spectral energy). To reduce redundancy I applied Principal Component Analysis (PCA) and kept the top 10 components. Top Contributing Features to Each PCA Component:

PC	Main Contributors
1	Acc_Z_fft_mean, Acc_Z_std, Gyro_Y_fft_mean, Gyro_X_fft_mean
2	Acc_X_spectral_energy, Acc_X_fft_std, Acc_Y_fft_std
3	Gyro_Z_std, Gyro_Z_fft_std, Gyro_Z_MAD
4	Acc_X_MAD, Gyro_Z_dom_freq, Gyro_X_dom_freq
5	Acc_Y_min, Acc_Y_mean, Acc_Y_std
6-10	Higher-order skewness / kurtosis from Acc and Gyro channels

These features summarize motion magnitude, variation, and rhythm—key traits of an individual's gait.

Architecture and Training

- Input = 10 PCA features $\rightarrow$ two hidden layers (5 neurons each, ReLU) $\rightarrow$ output (10 Softmax).

2. Optimizer = Adam, Loss = categorical cross-entropy.
3. Five-fold stratified cross-validation (30 epochs, early stopping = 3).

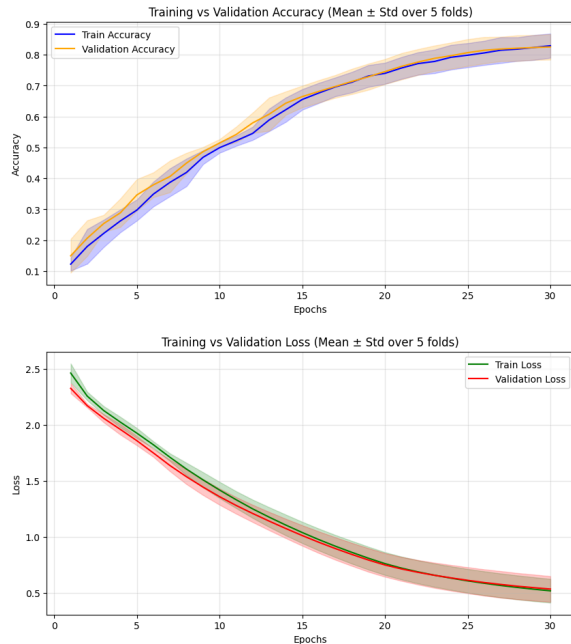


Figure 2: Training vs Validation Accuracy/Loss

2.3 LSTM Network for Identification

1. Input = 250 time steps $\times$ 6 features.
2. Architecture = five stacked LSTM layers (6 units each, dropout 0.3 + batch norm between layers),
3. dense(6, ReLU) + Softmax(10).
4. Trained with Adam optimizer, binary cross-entropy loss, 35 epochs
5. Cross-validated with 5 folds.

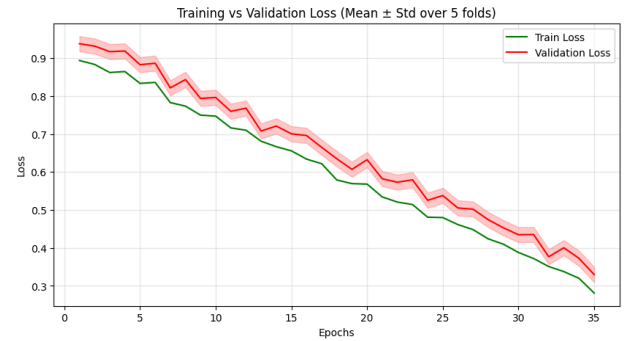
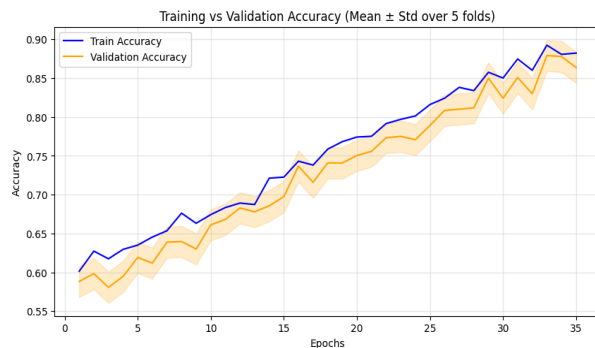


Figure 3: LSTM Training/Validation Curves

3. Results

3.1 Identification Results

Model	Accuracy	Precision	Recall	F1-Score
FNN	0.90	0.90	0.89	0.89
LSTM	0.57	0.64	0.54	0.47

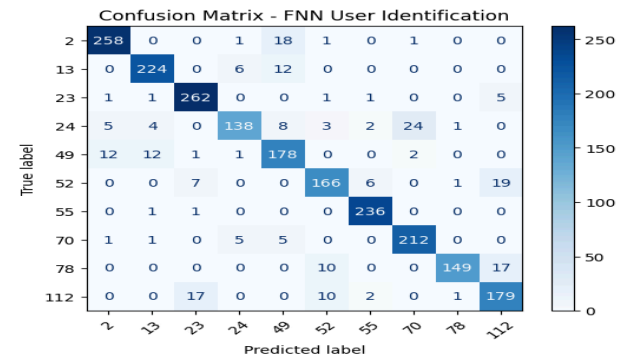


Figure 4: FNN Confusion Matrix

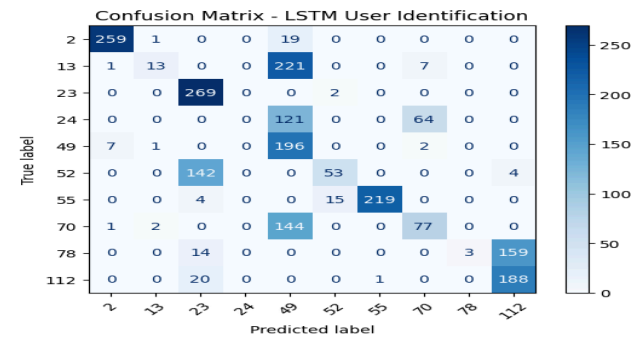


Figure 5: LSTM Confusion Matrix

3.2 Verification Results

Model	Accuracy	AUC	FAR	FRR	EER
FNN	0.97	0.99	0.12	0.25	0.03
LSTM	0.97	0.98	0.16	0.36	0.02

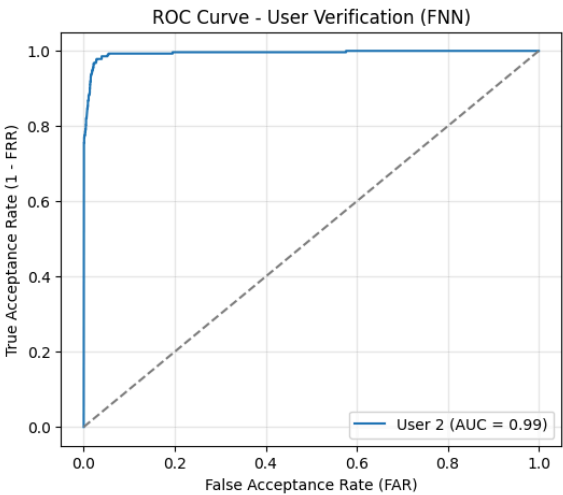


Figure 6: ROC Curve for One User — FNN

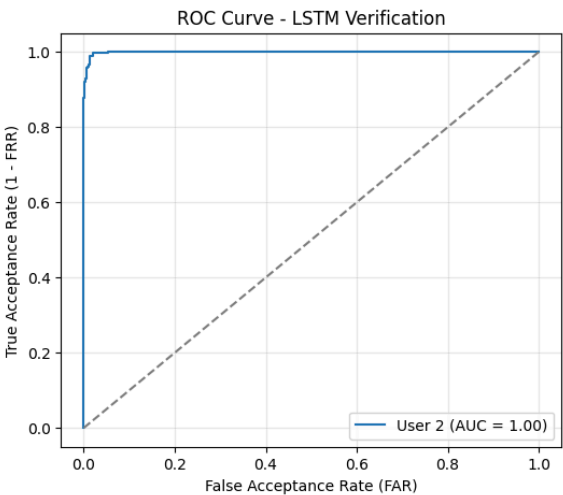


Figure 6: ROC Curve for One User — LSTM

4. Discussion and Insights

The Feedforward Neural Network (FNN) The Feedforward Neural Network (FNN) generally performed better and more consistently than the Long Short-Term Memory

(LSTM) model in both identification and verification. The FNN benefited from using handcrafted statistical and frequency features that captured clear gait patterns even with limited data, while the LSTM had to learn these patterns directly from the raw signals and often struggled due to the small dataset and restricted network size. Verification tasks were easier for both models than identification since each verifier compared only one user against others, making the decision boundary simpler. Training the LSTM models took longer and required careful tuning of normalization and early stopping, while the FNN trained faster and was easier to stabilize. I also faced challenges with data imbalance and overlapping gait patterns between users, which affected model precision for certain individuals. Overall, the FNN was more efficient, stable, and suitable for smaller datasets, while the LSTM showed potential for improvement with larger data and longer sequences.

5. Conclusion

This project explored gait-based biometric identification and verification using accelerometer and gyroscope data. I implemented two neural network approaches, an FNN using extracted features and an LSTM using raw signals. After careful preprocessing with outlier removal and smoothing, both models were able to learn user-specific walking patterns, with the FNN providing stronger and more stable results. Verification proved to be more reliable than identification for both architectures. Overall, the experiments show that well-engineered features combined with simple neural networks can achieve dependable performance in gait-based biometric systems, and sequence models like LSTM could perform better with larger, more diverse datasets.